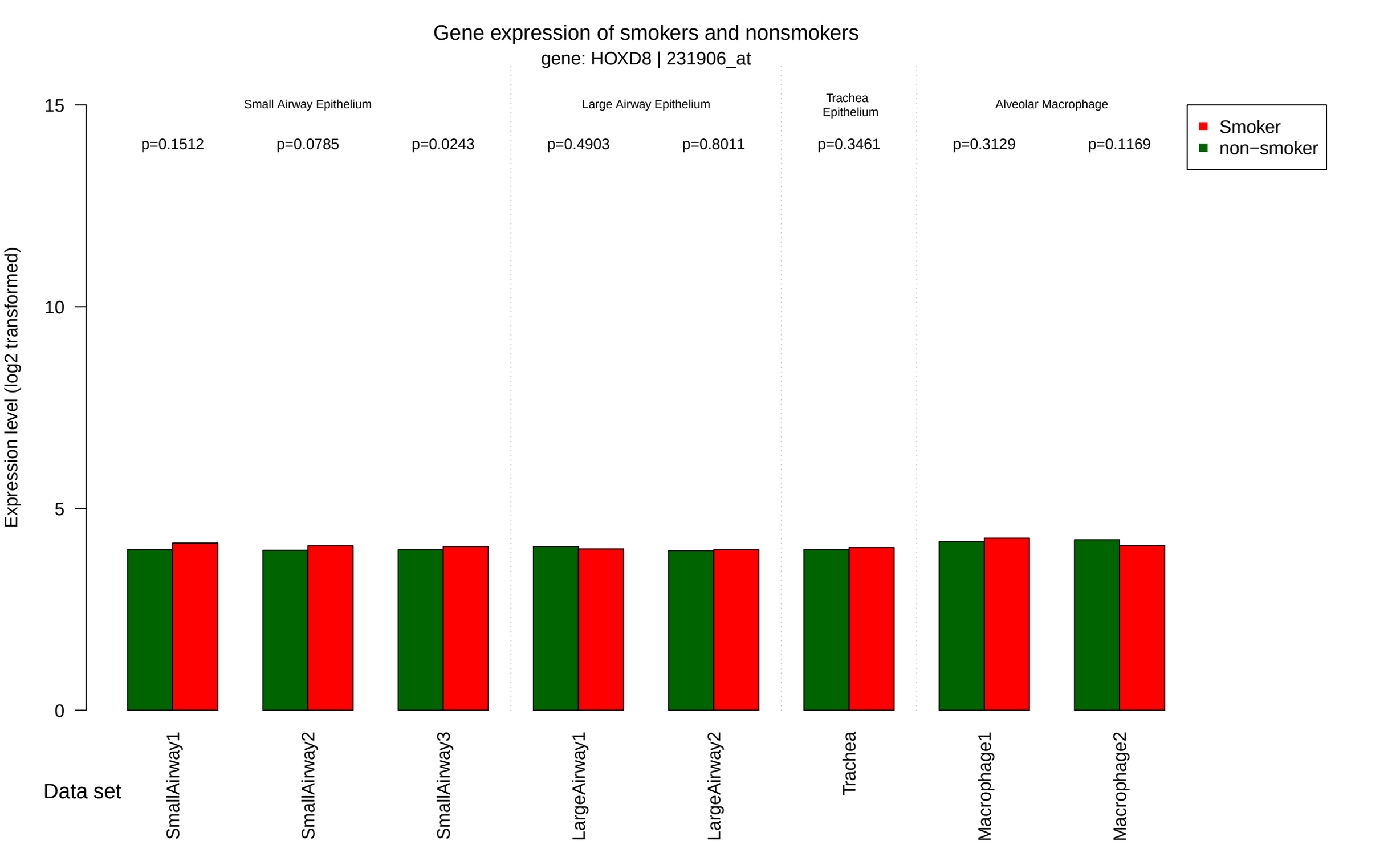
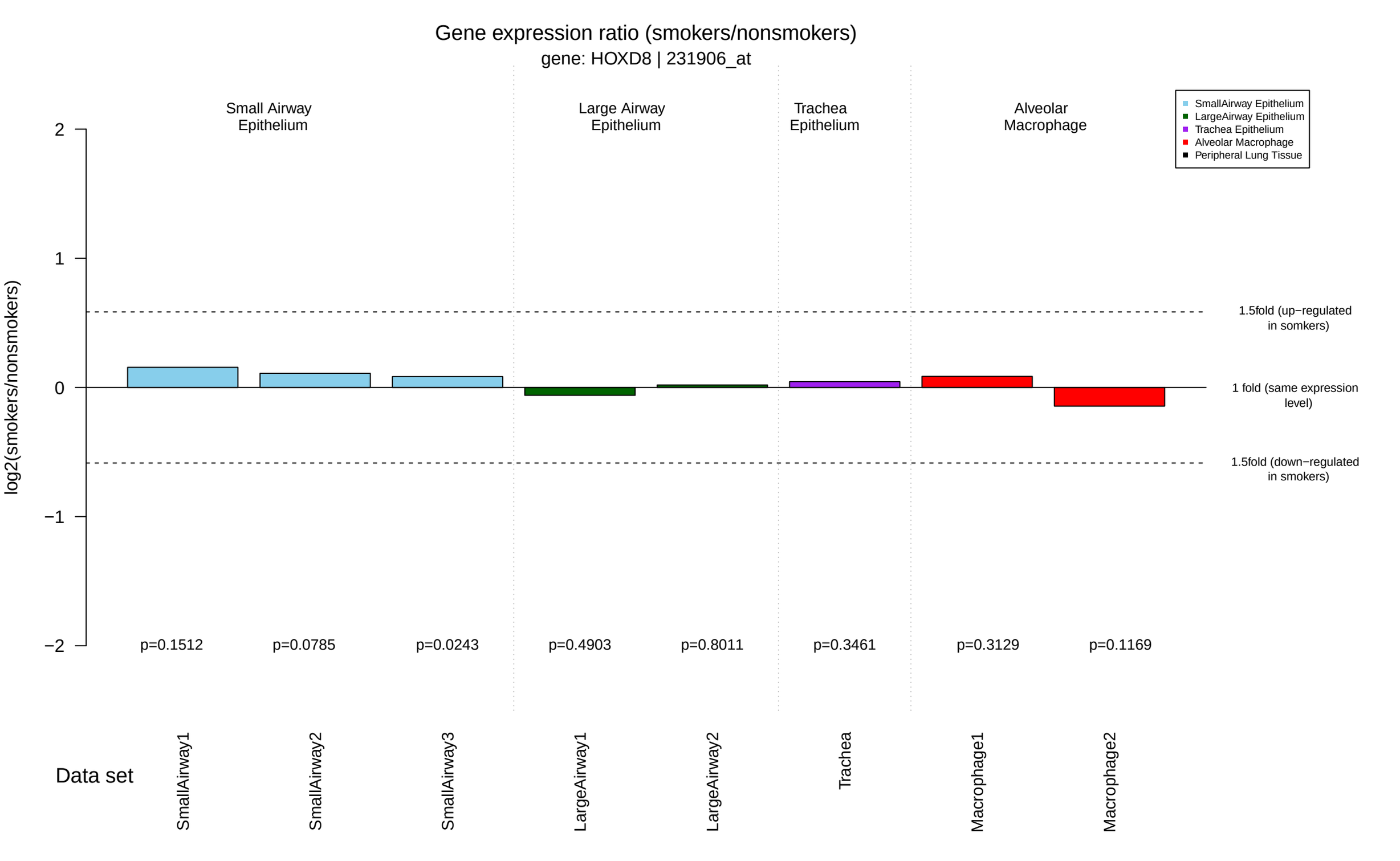
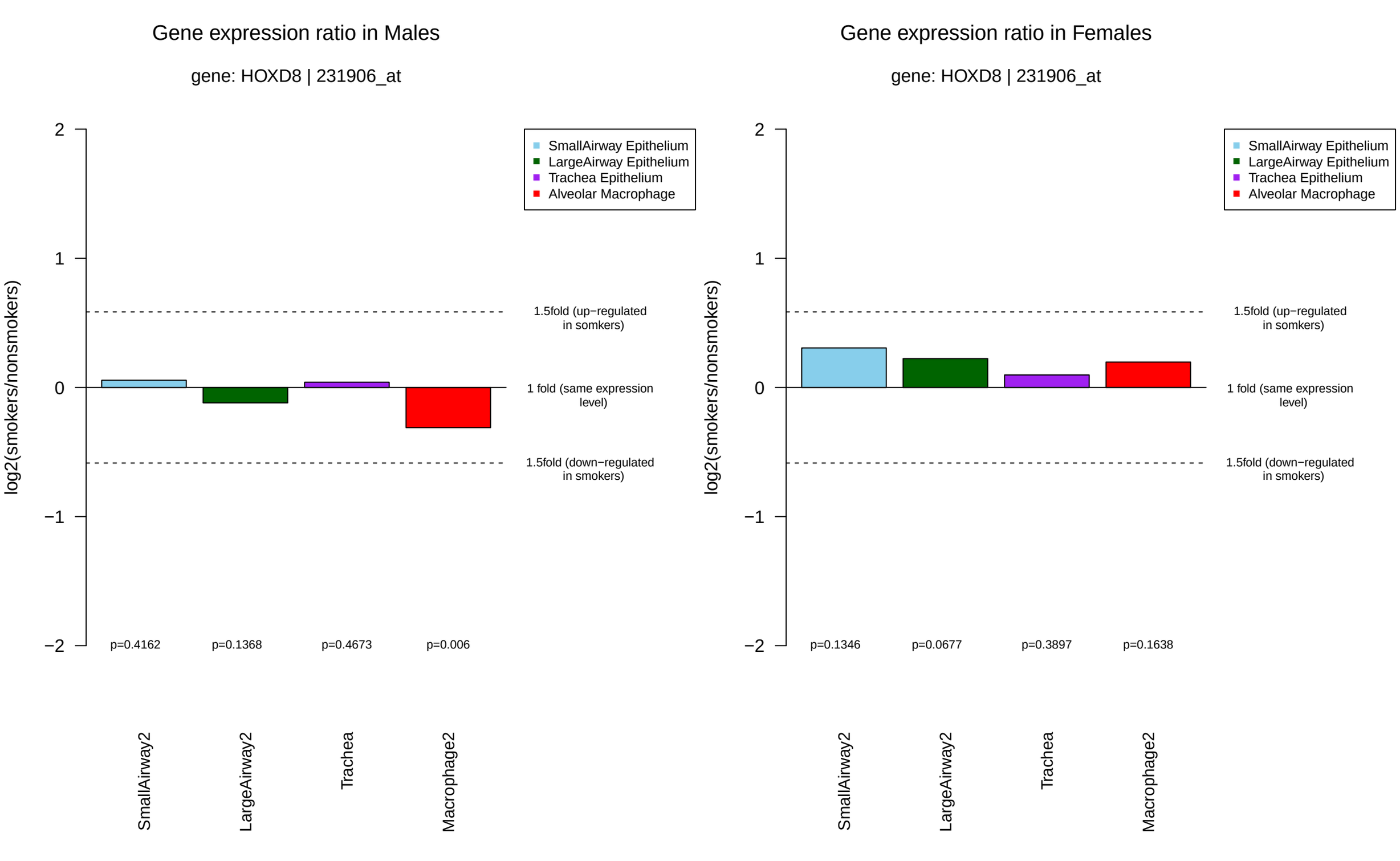
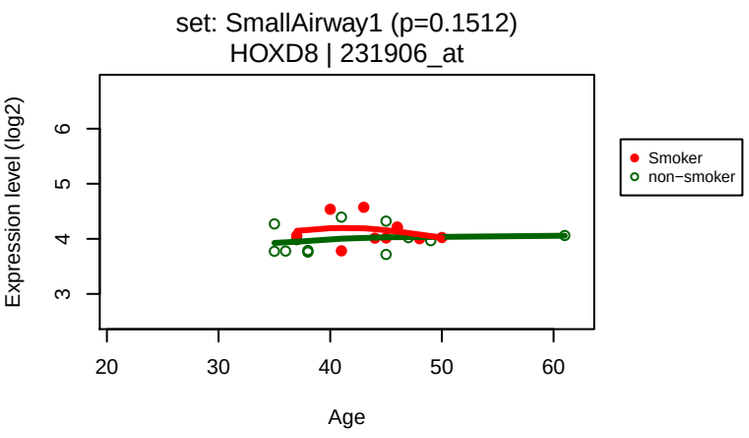
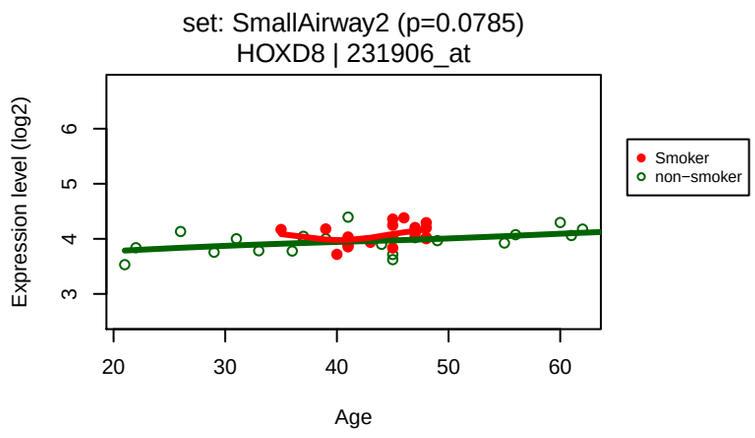
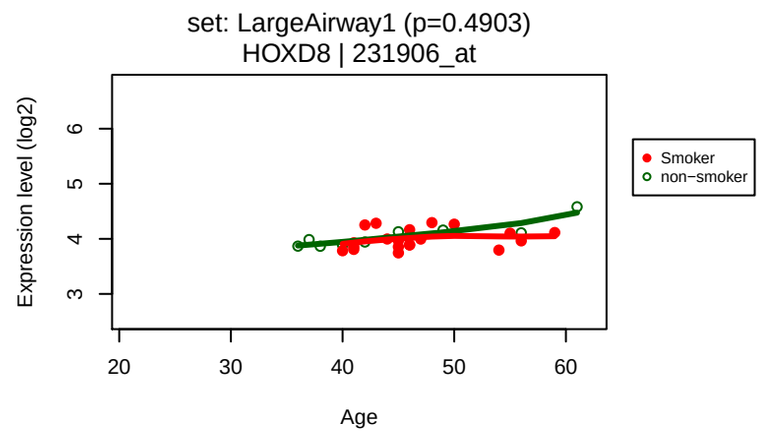

Gene expression of smokers and nonsmokers
gene: HOXD8 | 231906_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1512
p=0.0785
p=0.0243
p=0.4903
p=0.8011
p=0.3461
p=0.3129
p=0.1169
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

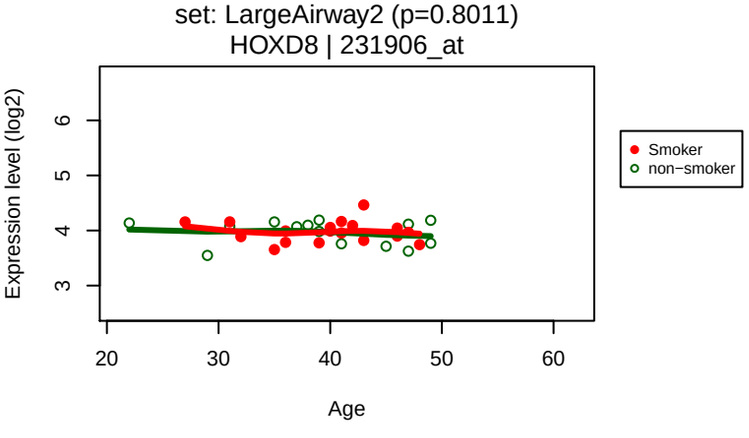

Gene expression ratio (smokers/nonsmokers)
gene: HOXD8 | 231906_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1512
p=0.0785
p=0.0243
p=0.4903
p=0.8011
p=0.3461
p=0.3129
p=0.1169
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

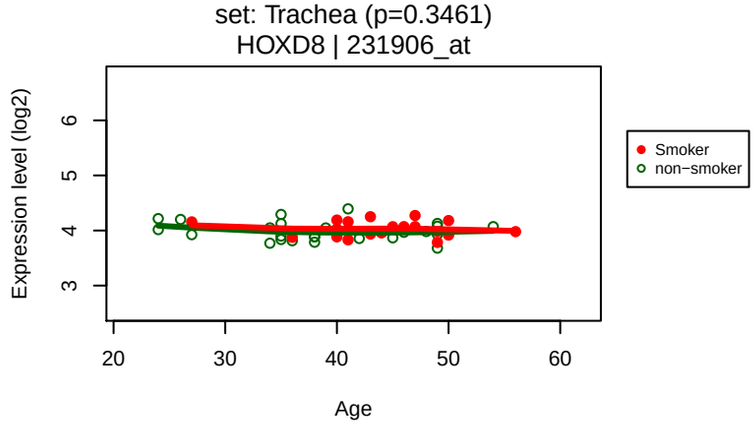

Gene expression ratio in Males
gene: HOXD8 | 231906_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4162
p=0.1368
p=0.4673
p=0.006
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXD8 | 231906_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1346
p=0.0677
p=0.3897
p=0.1638
SmallAirway2
LargeAirway2
Trachea
Macrophage2

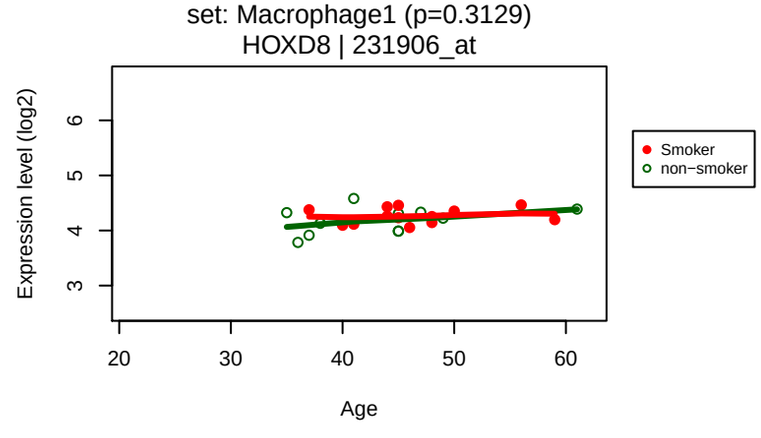

set: SmallAirway1 (p=0.1512)
HOXD8 | 231906_at
Expression level (log2)
Age
Smoker
non−smoker

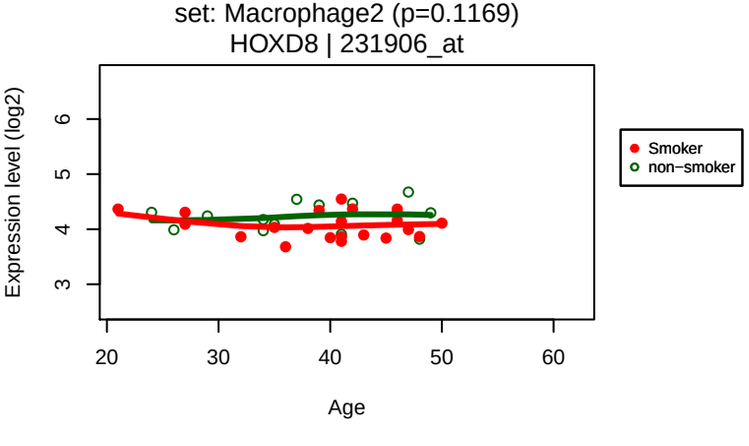

set: SmallAirway2 (p=0.0785)
HOXD8 | 231906_at
Expression level (log2)
Age
Smoker
non−smoker

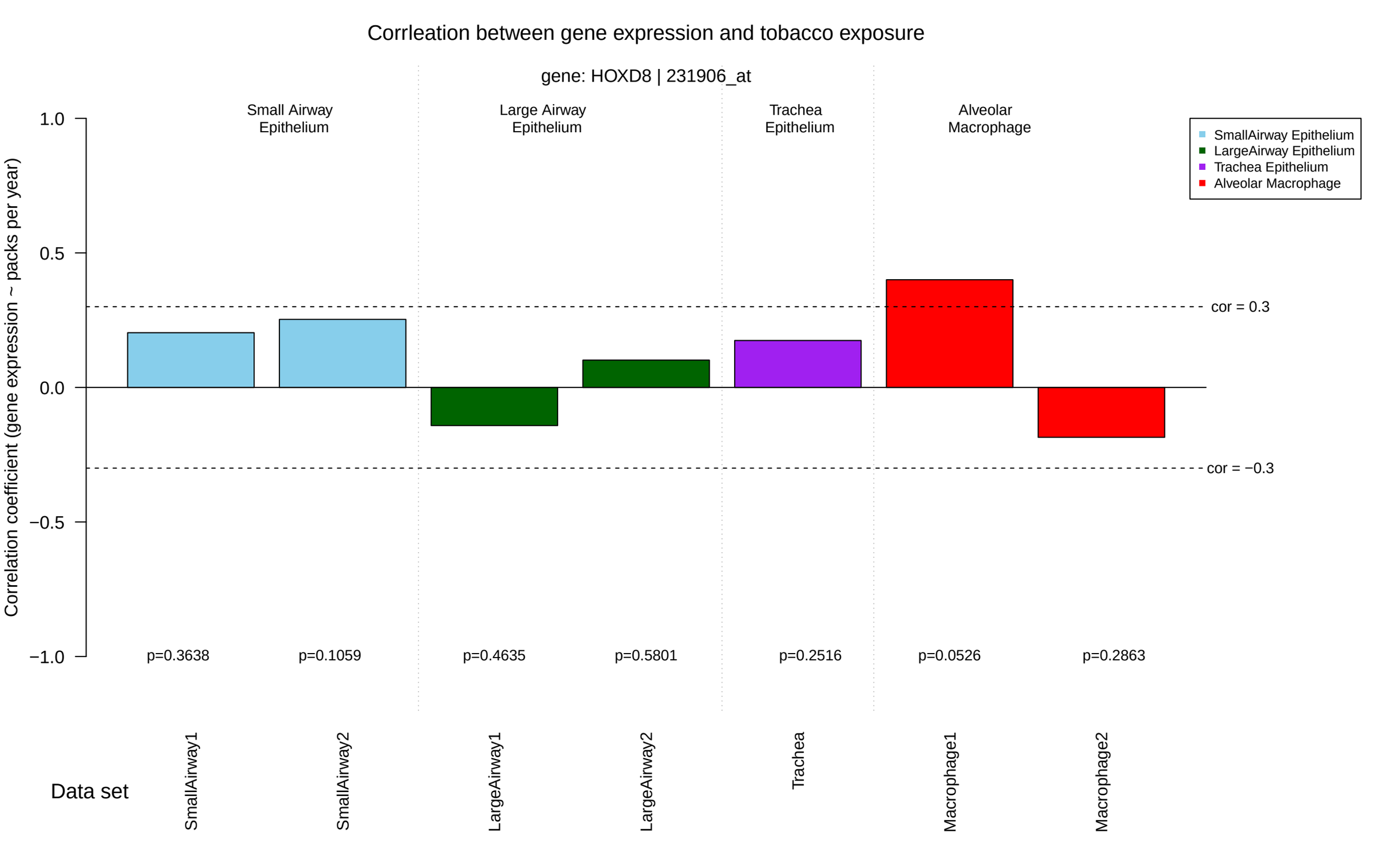

set: LargeAirway1 (p=0.4903)
HOXD8 | 231906_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8011)
HOXD8 | 231906_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3461)
HOXD8 | 231906_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3129)
HOXD8 | 231906_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1169)
HOXD8 | 231906_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXD8 | 231906_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3638
p=0.1059
p=0.4635
p=0.5801
p=0.2516
p=0.0526
p=0.2863
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2